

● HARD

● ADVANCED MATH

● ANSWER KEY

Advanced Math

01

10 answers · Rationale-rich · Teach from doc

Q1**336**

The correct answer is 336. By the zero product property, if $x + 14t - x = 0$, then $x + 14 = 0$, which gives $x = -14$, or $t - x = 0$, which gives $x = t$. Therefore, $gx = 0$ when $x = -14$ and when $x = t$. Since the graph of $y = gx$ passes through the point $(24, 0)$, it follows that $g(24) = 0$, so $t = 24$. Substituting 24 for t in the equation $gx = x + 14t - x$ yields $gx = x + 14(24) - x$. The value of $g(0)$ can be calculated by substituting 0 for x in this equation, which yields $g(0) = 0 + 14(24) - 0$, or $g(0) = 336$.

Q2**D****D. 1**

Choice D is correct. The system of equations can be solved using the substitution method. Solving the second equation for y gives $y = -x - 1$. Substituting the expression $-x - 1$ for y into the first equation gives $-x - 1 = x^2 + 2x + 1$. Adding $x + 1$ to both sides of the equation yields $x^2 + 3x + 2 = 0$. The left-hand side of the equation can be factored by finding two numbers whose sum is 3 and whose product is 2, which gives $(x + 2)(x + 1) = 0$. Setting each factor equal to 0 yields $x + 2 = 0$ and $x + 1 = 0$, and solving for x yields $x = -2$ or $x = -1$. These values of x can be substituted for x in the equation $y = -x - 1$ to find the corresponding y -values: $y = -(-2) - 1 = 2 - 1 = 1$ and $y = -(-1) - 1 = 1 - 1 = 0$. It follows that $(-2, 1)$ and $(-1, 0)$ are the solutions to the given system of equations. Therefore, $(x_1, y_1) = (-2, 1)$, $(x_2, y_2) = (-1, 0)$, and $y_1 + y_2 = 1 + 0 = 1$.

Choice A is incorrect. The solutions to the system of equations are $(x_1, y_1) = (-2, 1)$ and $(x_2, y_2) = (-1, 0)$. Therefore, -3 is the sum of the x -coordinates of the solutions, not the sum of the y -coordinates of the solutions. Choices B and C are incorrect and may be the result of computation or substitution errors.

Q3**D**

D. $\frac{42ak^2 + 1}{k}$

Choice D is correct. Two fractions can be added together when they have a common denominator. Since $k > 0$, multiplying the second term in the given expression by $\frac{k}{k}$ yields $\frac{42akk}{k}$, which is equivalent to $\frac{42ak^2}{k}$. Therefore, the expression $\frac{42a}{k} + 42ak$ can be written as $\frac{42a}{k} + \frac{42ak^2}{k}$ which is equivalent to $\frac{42a + 42ak^2}{k}$. Since each term in the numerator of this expression has a factor of $42a$, the expression $\frac{42a + 42ak^2}{k}$ can be rewritten as $\frac{42a(1 + k^2)}{k}$, or $\frac{42a(1 + k^2)}{k}$, which is equivalent to $\frac{42a(k^2 + 1)}{k}$.

Choice A is incorrect. This expression is equivalent to $\frac{42a}{k} + \frac{42a}{k}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $\frac{42a}{k} + 42a$.

Q4

D

D. -12

Choice D is correct. The equation of a parabola in the xy -plane can be written in the form $y = ax^2 - h^2 + k$, where a is a constant and h, k is the vertex of the parabola. If a is positive, the parabola will open upward, and if a is negative, the parabola will open downward. It's given that the parabola has vertex $9, -14$. Substituting 9 for h and -14 for k in the equation $y = ax^2 - h^2 + k$ gives $y = ax^2 - 9^2 - 14$, which can be rewritten as $y = ax^2 - 18x + 81 - 14$, or $y = ax^2 - 18ax + 81a - 14$. Distributing the factor of a on the right-hand side of this equation yields $y = ax^2 - 18ax + 81a - 14$. Therefore, the equation of the parabola, $y = ax^2 - 18ax + 81a - 14$, can be written in the form $y = ax^2 + bx + c$, where $a = a$, $b = -18a$, and $c = 81a - 14$. Substituting $-18a$ for b and $81a - 14$ for c in the expression $a + b + c$ yields $a + -18a + 81a - 14$, or $64a - 14$. Since the vertex of the parabola, $9, -14$, is below the x -axis, and it's given that the parabola intersects the x -axis at two points, the parabola must open upward. Therefore, the constant a must have a positive value. Setting the expression $64a - 14$ equal to the value in choice D yields $64a - 14 = -12$. Adding 14 to both sides of this equation yields $64a = 2$. Dividing both sides of this equation by 64 yields $a = \frac{2}{64}$, which is a positive value. Therefore, if the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants, the value of $a + b + c$ could be -12 .

Choice A is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants and $a + b + c = -23$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the x -axis at zero points, not two points.

Choice B is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants and $a + b + c = -19$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the x -axis at zero points, not two points.

Choice C is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants and $a + b + c = -14$, then the value of a will be 0 , which is inconsistent with the equation of a parabola.

Q5

B

B. $\{5\}$

Choice B is correct. Subtracting 4 from both sides of $\sqrt{2x+6} + 4 = x + 3$ isolates the radical expression on the left side of the equation as follows: $\sqrt{2x+6} = x - 1$. Squaring both sides of $\sqrt{2x+6} = x - 1$ yields $2x + 6 = x^2 - 2x + 1$. This equation can be rewritten as a quadratic equation in standard form: $x^2 - 4x - 5 = 0$. One way to solve this quadratic equation is to factor the expression $x^2 - 4x - 5$ by identifying two numbers with a sum of -4 and a product of -5 . These numbers are -5 and 1. So the quadratic equation can be factored as $(x - 5)(x + 1) = 0$. It follows that 5 and -1 are the solutions to the quadratic equation. However, the solutions must be verified by checking whether 5 and -1 satisfy the original equation, $\sqrt{2x+6} + 4 = x + 3$. When $x = -1$, the original equation gives $\sqrt{2(-1)+6} + 4 = (-1) + 3$, or $6 = 2$, which is false. Therefore, -1 does not satisfy the original equation. When $x = 5$, the original equation gives $\sqrt{2(5)+6} + 4 = 5 + 3$, or $8 = 8$, which is true. Therefore, $x = 5$ is the only solution to the original equation, and so the solution set is $\{5\}$.

Choices A, C, and D are incorrect because each of these sets contains at least one value that results in a false statement when substituted into the given equation. For instance, in choice D, when 0 is substituted for x into the given equation, the result is $\sqrt{2(0)+6} + 4 = (0) + 3$, or $\sqrt{6} + 4 = 3$. This is not a true statement, so 0 is not a solution to the given equation.

Q6**D**D. $\frac{45}{k}$

Choice D is correct. It's given that $4x^2 + bx - 45$ can be rewritten as $hx + kx + j$. The expression $hx + kx + j$ can be rewritten as $hx^2 + jhx + kx + kj$, or $hx^2 + jh + kx + kj$. Therefore, $hx^2 + jh + kx + kj$ is equivalent to $4x^2 + bx - 45$. It follows that $kj = -45$. Dividing each side of this equation by k yields $j = \frac{-45}{k}$. Since j is an integer, $\frac{-45}{k}$ must be an integer. Therefore, $\frac{45}{k}$ must also be an integer.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7**-324**

The correct answer is -324. It's given that the function f is defined by $fx = a^x + b$, where a and b are constants. It's also given that the graph of $y = fx$ has a y -intercept at $0, -323$. It follows that $f0 = -323$. Substituting 0 for x and -323 for fx in $fx = a^x + b$ yields $-323 = a^0 + b$, or $-323 = 1 + b$. Subtracting 1 from each side of this equation yields $-324 = b$. Therefore, the value of b is -324 .

Q8**289**

The correct answer is 289. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solutions when the value of the discriminant, $b^2 - 4ac$, is less than 0 . In the given equation, $x^2 - 34x + c = 0$, $a = 1$ and $b = -34$. Therefore, the discriminant of the given equation can be expressed as $-34^2 - 4(1)c$, or $1,156 - 4c$. It follows that the given equation has no real solutions when $1,156 - 4c < 0$. Adding $4c$ to both sides of this inequality yields $1,156 < 4c$. Dividing both sides of this inequality by 4 yields $289 < c$, or $c > 289$. It's given that the equation $x^2 - 34x + c = 0$ has no real solutions when $c > n$. Therefore, the least possible value of n is 289 .

Q9**.09**

Accept also: 9/100

The correct answer is .09 . It's given that the expression $0.36x^2 + 0.63x + 1.17$ can be rewritten as $ax^2 + 7x + 13$. Applying the distributive property to the expression $ax^2 + 7x + 13$ yields $4ax^2 + 7ax + 13a$. Therefore, $0.36x^2 + 0.63x + 1.17$ can be rewritten as $4ax^2 + 7ax + 13a$. It follows that in the expressions $0.36x^2 + 0.63x + 1.17$ and $4ax^2 + 7ax + 13a$, the coefficients of x^2 are equivalent, the coefficients of x are equivalent, and the constant terms are equivalent. Therefore, $0.36 = 4a$, $0.63 = 7a$, and $1.17 = 13a$. Solving any of these equations for a yields the value of a . Dividing both sides of the equation $0.36 = 4a$ by 4 yields $0.09 = a$. Therefore, the value of a is 0.09. Note that .09 and 9/100 are examples of ways to enter a correct answer.

Q10**210**

The correct answer is 210. Since -5 , 6, and 7 are zeros of the function, the function can be rewritten as $h(x) = (x+5)(x-6)(x-7)$. Expanding the function yields $h(x) = x^3 - 8x^2 - 23x + 210$. Thus, $a = -8$, $b = -23$, and $c = 210$. Therefore, the value of c is 210.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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